

Time-domain Simulation of Battery Warburg Impedance via RC-Ladder Approximation

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Abstract—The Warburg impedance in the frequency domain models the low-frequency diffusion behavior of a lithium-ion battery, but its fractional order gives it no exact finite-dimensional time-domain realization, and no consensus exists on which finite-order approximation and time-domain integration scheme best suits real-time battery simulation. This paper fits an N -branch RC ladder network to the ideal Warburg impedance via nonlinear least squares, selects the ladder order using RMSE and the Bayesian Information Criterion, and compares three time-domain integration schemes – zero-order-hold, forward Euler, and an interpolated fourth-order Runge–Kutta method – for simulating the resulting network under arbitrary current excitation.

Index Terms—lithium-ion battery, battery management systems, electrochemical impedance spectroscopy, Warburg impedance, equivalent-circuit model, RC ladder network, numerical integration

I. PROBLEM FORMULATION

Accurate, real-time knowledge of a lithium-ion cell's State of Charge (SoC) and State of Health (SoH) is fundamental to safe and efficient operation of a Battery Management System (BMS). Both quantities are inferred from an underlying electrical model of the cell, so the fidelity of that model directly limits how tightly a BMS can guard against overcharge, over-discharge, and accelerated capacity fade while still exploiting the full usable energy of the pack. Equivalent-circuit models remain the dominant choice for this task precisely because SoC/SoH estimation must execute in real time on embedded, resource-constrained automotive microcontrollers, making the trade-off between model accuracy and computational cost a central design constraint.

Of the physical processes an equivalent-circuit model must capture, solid-state diffusion of lithium ions within the electrode particles dominates the cell's impedance response at low excitation frequencies and is classically represented by the Warburg impedance. Unlike the resistive and capacitive elements of a Randles-type equivalent circuit, the Warburg element is fractional-order: it scales as $1/\sqrt{j\omega}$ rather than as

an integer power of frequency, which is precisely what gives a diffusion-limited cell its characteristic low-frequency Nyquist tail.

This fractional order is also what makes the Warburg element difficult to simulate in the time domain. In the Laplace domain it corresponds to an $s^{-0.5}$ operator – a half-order integrator with no finite-dimensional state-space realization – and evaluating it exactly requires a fractional-calculus operation whose output at any instant depends on the entire past history of the input. Such an operation is impractical to compute step-by-step on the fixed-point, resource-constrained microcontrollers used in real-time BMS applications, so any deployable implementation must instead replace the exact Warburg impedance with a finite-order, integer-order approximation that a microcontroller can integrate incrementally.

In the frequency domain, the Warburg impedance is given by

$$Z_W(\omega) = \frac{A_W}{\sqrt{j\omega}} \quad (1)$$

where A_W is the Warburg coefficient and ω is the angular frequency. Because Z_W has no finite-order rational form, it cannot be realized as a lumped circuit or driven directly in the time domain. In this paper, we address this constraint by approximating Z_W with an N -branch RC ladder network whose parameters are obtained through nonlinear curve fitting against the ideal Warburg response, and we compare three time-domain integration schemes for simulating the resulting network under an arbitrary current.

A. The Randles Equivalent Circuit

Because the Warburg element rarely appears in isolation in practice, it is useful to first place it within the classical Randles equivalent circuit, which is the standard qualitative picture used throughout electrochemical impedance spectroscopy (EIS) and the context in which the rest of this paper's results should be read. Fig. 1 shows the resulting Nyquist plot for assumed, known values of the non-Warburg elements,

comparing it against the fitted $N = 4$ RC ladder introduced in Section II.

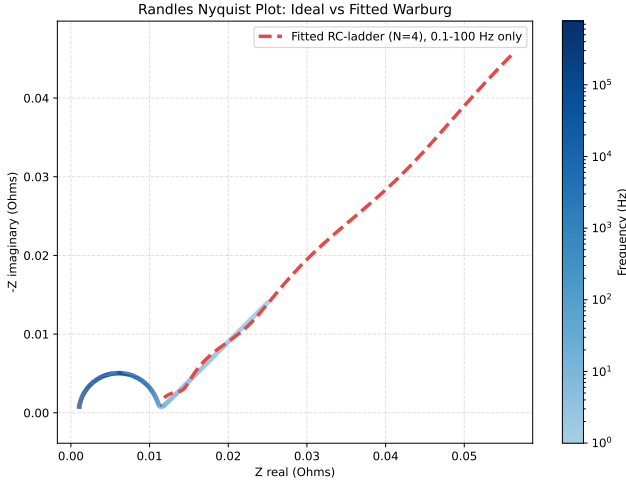


Fig. 1. Randles Nyquist plot using assumed R_s , R_{ct} , C_{dl} values, comparing the ideal analytic Warburg impedance (color-coded by frequency) to the fitted $N = 4$ RC ladder (dashed red) over the 0.1–100 Hz band it was fit on.

This shape is produced by the circuit shown in Fig. 2: a series resistance R_s , in series with a parallel combination of the double-layer capacitance C_{dl} and a branch containing the charge-transfer resistance R_{ct} in series with the Warburg impedance Z_W .

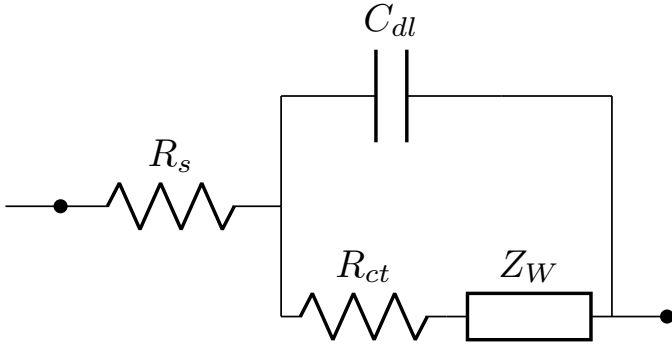


Fig. 2. The Randles equivalent circuit.

Each element represents a distinct physical process inside the cell, and each dominates the measured impedance over a different frequency range, visible as a corresponding feature in Fig. 1. R_s is the ohmic (series) resistance of the electrolyte, current collectors, and contacts; having no frequency dependence, it simply shifts the entire spectrum along the real axis and sets the high-frequency intercept. R_{ct} and C_{dl} together represent the charge-transfer kinetics at the electrode-electrolyte interface and the electrical double layer that forms there; in parallel, they produce the semicircle visible in the mid-frequency range. Z_W , the Warburg impedance and the focus of this paper, represents solid-state diffusion of lithium ions within the electrode particles, and it is only the dominant

contributor to the total impedance at low frequency, appearing as the diagonal tail at the low-frequency end of the spectrum – at higher frequencies C_{dl} increasingly short-circuits the branch containing it. For the assumed values used in Fig. 1 ($R_s = 1 \text{ m}\Omega$, $R_{ct} = 10 \text{ m}\Omega$, $C_{dl} = 400 \text{ }\mu\text{F}$), Z_W dominates its branch only below approximately 4 Hz, and that branch is increasingly bypassed by C_{dl} above approximately 40 kHz.

These values are illustrative only, chosen to produce a clearly resolved semicircle and diffusion tail in Fig. 1 rather than measured from a specific cell; because the crossover frequency scales as $(A_W/R_{ct})^2$, it is highly sensitive to the assumed R_{ct} and is not an intrinsic property of the Warburg element itself. This illustrative crossover is independent of the 0.1–100 Hz band used to fit and validate the RC ladder throughout the remainder of this paper, which treats the Warburg element on its own terms without reference to R_{ct} or C_{dl} . This has a direct consequence for the RC-ladder approximation developed in the remainder of this paper: the ladder’s accuracy inherently degrades toward the high-frequency edge of the swept band (Fig. 4), but since Z_W itself is only physically relevant at low frequency once embedded in a full Randles circuit, this degradation is of limited practical consequence, as discussed further in Section II-A0c.

II. METHODOLOGY

A. RC Ladder Network

We approximate the Warburg element with N parallel $R_n C_n$ branches driven by a common current $i(t)$, with branch voltages summed:

$$Z_{RC}(\omega) \approx \sum_{n=1}^N \frac{R_n}{1 + j\omega R_n C_n}. \quad (2)$$

Rather than assigning the branch parameters $\{R_n, C_n\}$ from a closed-form scaling law, we solve for them directly: the $2N$ -parameter vector $\theta = (R_1, \dots, R_N, C_1, \dots, C_N)$ is obtained by nonlinear least squares so that the ladder’s frequency response (2) matches the ideal Warburg impedance (1) across the frequency sweep $\{\omega_m\}_{m=1}^M$ used later for extraction (Section II-C):

$$\theta^* = \arg \min_{\theta \geq 0} \sum_{m=1}^M |Z_{RC}(\omega_m; \theta) - Z_W(\omega_m)|^2. \quad (3)$$

The optimization is bounded ($R_n, C_n \geq 0$) and solved with a nonlinear least-squares routine (`scipy.optimize.curve_fit`), seeded from a log-spaced initial guess: time constants

$$\tau_n^{(0)} = \tau_{\min} \left(\frac{\tau_{\max}}{\tau_{\min}} \right)^{\frac{n-1}{N-1}}, \quad n = 1, \dots, N, \quad (4)$$

with initial values $R_n^{(0)} = A_W \sqrt{\tau_n^{(0)}}$, $C_n^{(0)} = \tau_n^{(0)} / R_n^{(0)}$ — the same $\sqrt{\tau}$ scaling that makes an unoptimized ladder trace an approximate 45° Nyquist line. Fitting (3) then refines this starting point into the branch values that minimize impedance error over the actual sweep band, rather than relying on the

asymptotic argument that the scaling law only holds as $N \rightarrow \infty$ with densely spaced τ_n .

a) *Model order selection* ($N = 4$): To choose N , we refit (3) for $N \in \{2, \dots, 8\}$ and track the resulting RMS impedance error, $\text{RMSE}(N) = \sqrt{\frac{1}{M} \sum_m |Z_{RC}(\omega_m; \theta^*) - Z_W(\omega_m)|^2}$, alongside the Bayesian Information Criterion evaluated on the stacked real/imaginary residual (Fig. 3). $\text{RMSE}(N)$ falls from 2.9 m Ω at $N = 2$ to 0.44 m Ω at $N = 4$, after which each additional branch pair buys a much smaller absolute reduction — only a further 0.43 m Ω combined across $N = 5$ through $N = 8$. We adopt $N = 4$ as a practical working order for this stage of the study: the fit error is already well below any expected experimental noise floor, the branch count stays small enough for straightforward time-domain implementation, and the optimizer still returns four distinct, monotonically ordered time constants spanning nearly four decades (2.8 ms to 4.8 s) that are physically interpretable as increasingly slow diffusion-like branches.

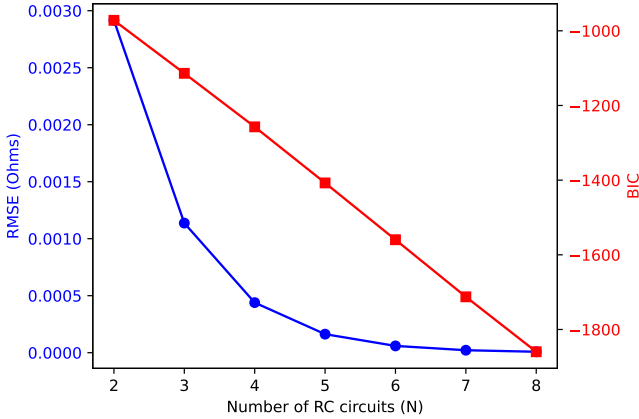


Fig. 3. RMSE and BIC of the fitted RC ladder as a function of the number of branches N . Absolute RMSE improvement per added branch shrinks markedly beyond $N = 4$.

Neither $\text{RMSE}(N)$ nor $\text{BIC}(N)$ level off within this range; this is expected rather than anomalous, since both are computed against the exact, noiseless analytic Warburg target rather than measured data. BIC's complexity penalty is calibrated to counteract fitting of random measurement noise, and no such noise floor exists here for the penalty to eventually outweigh — BIC is only expected to level off once applied to real, noisy experimental impedance data, which is left for future work.

Each branch obeys a first-order linear ODE driven by the same $i(t)$:

$$\frac{dv_n}{dt} = \frac{i(t) - v_n/R_n}{C_n}, \quad n = 1, \dots, N, \quad (5)$$

and the total ladder (Warburg) voltage is

$$v_W(t) = \sum_{n=1}^N v_n(t). \quad (6)$$

Because branches are coupled only through the shared forcing $i(t)$, (5) is integrated independently and in parallel for all n , using the fitted θ^* from (3).

b) *Frequency-dependent error validation*: A single scalar RMSE can mask error that is concentrated in a narrow part of the spectrum, so before passing θ^* to the time-domain solvers introduced next, we also evaluate the pointwise magnitude error

$$e(\omega_m) = \frac{|Z_W(\omega_m) - Z_{RC}(\omega_m; \theta^*)|}{|Z_W(\omega_m)|} \quad (7)$$

across the full swept band (Fig. 4). This confirms the $N = 4$ ladder holds a tight, well-behaved error tolerance over most of the frequency range of interest, rather than relying on an averaged metric alone that could obscure large local deviations near the edges of the sweep; error remains below 6% up to roughly 20 Hz before climbing toward the high-frequency edge of the band, where the excitation frequency exceeds the ladder's fastest fitted time constant.

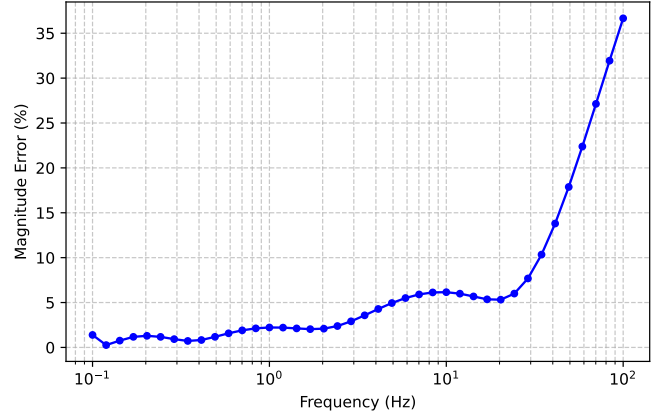


Fig. 4. Magnitude error between the ideal Warburg impedance and the fitted $N = 4$ RC ladder across the swept frequency band.

c) *Why the high-frequency error is inconsequential*:

The growing error toward the high-frequency edge of Fig. 4 is a genuine, structural limitation of a finite-order ladder in isolation: a single RC branch rolls off at -20 dB/decade past its own characteristic frequency, whereas the true Warburg impedance rolls off more gently, at -10 dB/decade (a consequence of the $\omega^{-0.5}$ power law), so once the excitation frequency exceeds the ladder's fastest fitted time constant it can no longer keep pace. As introduced in Section I-A, however, Z_W is only the dominant contributor to a full Randles circuit's impedance below its crossover frequency with R_{ct} (≈ 4 Hz for the assumed values in Fig. 1) — comfortably inside the region where the $N = 4$ ladder is most accurate. Above that crossover, the double-layer capacitance increasingly bypasses the branch containing Z_W entirely, so the ladder's growing error at the high-frequency edge of the swept band contributes negligibly to the impedance of the complete cell model.

B. Time-Domain Approximators

Given a sampled current $i[k] = i(kT_s)$ at step size T_s , we compare three ways of marching (5) forward — this is the actual point of comparison in this work, not the ladder topology itself.

1) *Approach 1: Zero-Order-Hold (ZOH)*: Assuming $i(t)$ held constant at $i[k+1]$ over each interval, (5) has a closed-form per-branch solution:

$$v_n[k+1] = e^{-T_s/\tau_n} v_n[k] + R_n \left(1 - e^{-T_s/\tau_n}\right) i[k+1]. \quad (8)$$

This is exact for the assumed piecewise-constant input at any T_s/τ_n ; its only error source is how well that ZOH assumption represents the true continuous current between samples.

2) *Approach 2: Forward Euler*:

$$v_n[k+1] = v_n[k] + T_s \frac{i[k] - v_n[k]/R_n}{C_n}. \quad (9)$$

First-order accurate ($O(T_s)$ local error); like (8), it uses only $i[k]$ over the whole step, with no mechanism for resolving how $i(t)$ varies within $[kT_s, (k+1)T_s]$.

3) *Approach 3: RK4 with Interpolated Input*: Classical fourth-order Runge–Kutta for the non-autonomous system (5) requires the forcing evaluated at four points within the step: t_k , $t_k + T_s/2$ (twice), and $t_k + T_s$. Since $i(t)$ is only known at $i[k]$ and $i[k+1]$, the midpoint value is obtained by linear interpolation,

$$i_{\text{mid}} = \frac{1}{2} (i[k] + i[k+1]), \quad (10)$$

giving stages

$$k_1 = f(v_n[k], i[k]), \quad (11)$$

$$k_2 = f\left(v_n[k] + \frac{T_s}{2} k_1, i_{\text{mid}}\right), \quad (12)$$

$$k_3 = f\left(v_n[k] + \frac{T_s}{2} k_2, i_{\text{mid}}\right), \quad (13)$$

$$k_4 = f(v_n[k] + T_s k_3, i[k+1]), \quad (14)$$

with $f(v, i) = (i - v/R_n)/C_n$, combined as

$$v_n[k+1] = v_n[k] + \frac{T_s}{6} (k_1 + 2k_2 + 2k_3 + k_4). \quad (15)$$

Unlike (8) and (9), this scheme does not treat $i(t)$ as frozen for the whole step: the interpolation in (10) tracks the continuously varying forcing within $[t_k, t_k + T_s]$ rather than approximating it by a single sample. This matters even for a clean, noise-free single-tone current $i(t) = \sin(\omega t)$ whenever T_s is not negligible relative to the drive period $1/f$: at the top of the swept band ($f = 100$ Hz, $T_s = 1$ ms, i.e. 10 samples per period), Approaches 1 and 2 accrue non-trivial error from holding $i[k]$ fixed across each step, while Approach 3's linear interpolation of the forcing keeps this discretization error an order of magnitude smaller. Because the current is otherwise a pure sinusoid, the resulting impedance-extraction error is attributable solely to the choice of time-domain integrator — a clean baseline discretization-error comparison, isolated from any input noise.

C. Frequency Response Extraction

For each drive frequency f , the ladder is excited with sampled current $i[k]$, the voltage $v_W[k]$ is produced by one of (8), (9), or (15), and the impedance is extracted after discarding the initial transient via a discrete Fourier transform at the bin nearest f :

$$\hat{Z}(f) = \frac{\text{FFT}\{v_W\}(f)}{\text{FFT}\{i\}(f)}. \quad (16)$$

Each approximator's accuracy is judged against the algebraic ladder impedance (2) evaluated at the fitted θ^* , which serves as ground truth independent of any time-domain discretization error.

III. RESULTS AND DISCUSSION

Figure 5 overlays the three time-domain approximators against the algebraic ladder impedance (2) on a Nyquist plot. At this scale the three trajectories are visually indistinguishable from the ground truth across the swept band; the differences between integrators only become apparent when the per-frequency error is examined directly.

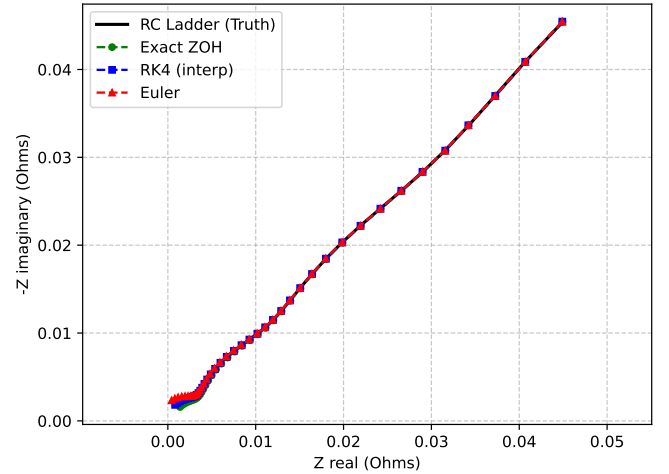


Fig. 5. Nyquist plot comparing the three time-domain integrators against the RC-ladder impedance evaluated algebraically from (2).

Figure 6 plots this impedance-extraction error on a log-log scale for all three integrators. The interpolated RK4 scheme (Approach 3) maintains the lowest error across nearly the entire band, staying below roughly 1% through most of the sweep and reaching only about 3% at the top of the band (100 Hz). The ZOH and forward Euler schemes, which both treat the drive current as frozen at a single sample for the whole step, grow past 1% error within the first decade and exceed 20% by 100 Hz, where only ten samples span each drive period. This is consistent with the discretization argument in Section II-C: for the fixed step size $T_s = 1$ ms used here, the dominant source of impedance-extraction error at high frequency is the choice of time-domain integrator rather than the RC-ladder approximation itself, which was already validated separately to sub-percent accuracy over most of the band (Fig. 4).

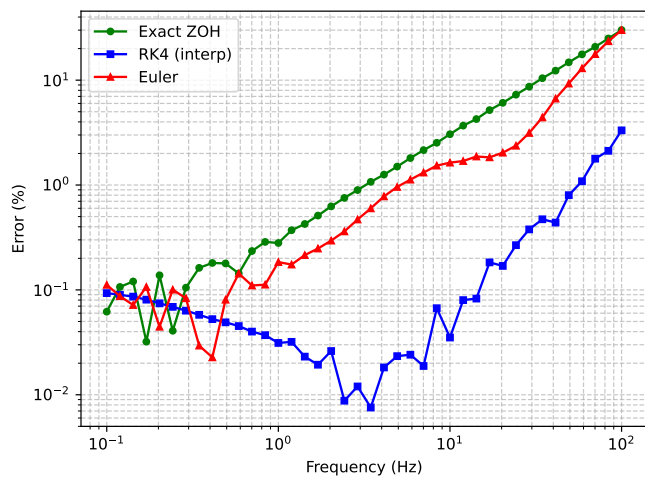


Fig. 6. Impedance-extraction error (%) versus frequency for the three time-domain integrators, relative to the algebraic RC-ladder impedance.